APAI Exam Instructions

AA. 2025- 2026

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- The exam will be conducted as a team project.
- Each group must consist of **a minimum of two – maximum of three members**.
- Individual projects are strictly prohibited due to the significant scale of the expected work.
- Start building your team **today!!!!**

- The project must be implemented using one of the programming languages introduced in the course (**TensorFlow**, **+/- Keras**, or **PyTorch**) and should address a problem through **deep learning methodologies**.
- **PROJECT FORMAT**
- The project must be delivered in the form of a scientific research paper.
  - This paper should have a link to the **code**, you can inject the URL of the code to the abstract of the paper and you should make the repo publicly available.
  - Your paper should be a ***minimum of 6 pages*** and a ***maximum of 8 pages*** **excluding references**.
  - A paper not obeying the page limits will be firmly rejected, thus the team has to re-write the project report or even repeat the project (this will be decided by the teacher).
  - As a template for the scientific research paper, you can use the CVPR 2025 templates given in
    - **https://github.com/cvpr-org/author-kit/releases**
      - The template can be either in Latex or in MS Word, *feel free to use one of the two you like*.
      - Suggestion: Referencing and injecting equations are significantly easier in the Latex template.
  - Thus, it is strongly recommended that teams investigate several accepted papers in the CVPR proceedings to better understand how a paper should be written (e.g., which sections exist and what is the purpose of each section).
    - The sections that are explicitly required are discussed in the following slides.
    - Examples for CVPR proceedings:
      - https://openaccess.thecvf.com/CVPR2024?day=all
      - https://openaccess.thecvf.com/CVPR2023?day=all
      - https://openaccess.thecvf.com/CVPR2022?day=all

- The paper should include the following as **<u>mandatory sections</u>**. *Not following the suggested agenda will result in a reduction in the score*.
  - **(1) Abstract**
    - A brief introduction to the problem and its significance.
    - A summary of the key challenges.
    - A concise description of the proposed approach.
    - A highlight of the main results and findings.
    - A statement on the impact or implications of the work.
  - **(2) Introduction**
    - Clearly define the problem being addressed (e.g., automatic anomaly detection in medical images).
    - Explain its significance (e.g., automatic methods can help medical doctors to prioritize the patient based on their emergency level).
      - Describe why the problem is important
      - Provide real-world applications (e.g., medical imaging, autonomous driving, surveillance).
    - Highlight the challenges that make it difficult (e.g., occlusions, variations in lighting, scalability, different sensors, lack of data…..).
    - Briefly summarize prior work in the field (literature review).
      - Highlight the limitations or gaps in existing solutions.
    - Clearly state the novel contributions of your approach.
      - Avoid just listing them; explain why they matter.
    - Provide a high-level summary of the proposed approach.
      - Keep details minimal—leave technical explanations for later sections.

- The paper should include the following as **<u>mandatory sections</u>**. *Not following the suggested agenda will result in a reduction in the score*.
  - **(3) Method**
    - A clear and formal description of the proposed approach.
    - A diagram illustrating the overall pipeline or architecture [VERY IMPORTANT].
    - Mathematical formulations (e.g., transformation, loss function).
    - Explanation of the deep learning architecture.
    - Details about datasets and preprocessing steps.
    - Implementation details, including training settings (e.g., optimizer, learning rate, hardware used).
    - Justification of design choices.
  - **(4) Results**
    - Comparison to baseline methods
      - You can create some baseline, more naïve methods to compare with your final proposed method. This is a good manner to also justify the effectiveness of your final proposed method with respect to some baselines.
    - Any ablation studies or variations of the final proposed method. [*Not compulsory but if realized then it will be very positively evaluated*]
      - Ablation study means evaluating the performance of the method when you change e.g., its parameters, its architecture such as a 1D CNN to LSTM and vice versa.
        - For example, instead of 3 layers if you have 2 layers how the performance is affected.
        - For example, if you change the number of augmentations, how the performance is affected.
      - You can supply the results in a table format or training/validation curves.

- The paper should include the following as **<u>mandatory sections</u>**. *Not following the suggested agenda will result in a reduction in the score*.
  - **(5) Conclusion**
    - A concise summary of the key findings.
    - Restate the main contributions without repeating the introduction verbatim.
    - Discuss the impact or significance of the results.
    - Highlight any limitations of the proposed approach.
    - Suggest possible directions for future work.

  - **(6) Contributions**
    - Each student’s contribution should be clearly detailed,
      - covering aspects such as coding, writing, idea development, literature review, and methodology.
      - If the teacher detects that a student has contributed less than stated and this was not disclosed, the entire team's grade may be affected.
      - This project is designed to encourage collaborative teamwork, so every student is expected to actively contribute.

- **How to Decide on the Project's Methodology**
  - *Some example methodologies* are listed below, but keep in mind that these topics have several variations (so there is no single approach to address them).
    - You can make a literature review and find a method (e.g., a paper on continual learning) you would like to implement and test it on several datasets.
    - Further, the team may also propose a new variation of these approaches with reasonings.
      - Such an approach can lead to a strong technical contribution and will be highly valued in the evaluation process.
    - Example topics (*cannot be methodologically simpler than these*)
      - Self-supervised learning (e.g., pre-training)
      - Knowledge distillation
      - Continual Learning
      - Meta Learning
      - VLMs / MLLMs (particularly CLIP, BLIP, CLIP with adapters, optimal transport, learnable prompts)

Once you have decided on **your methodology** and **dataset**, get confirmation from your teacher.

At this stage, you need to communicate with her, providing a description of the project (i.e., methodology) and the dataset(s) to be used.

If the dataset you want to use is too large, feel free to use a portion of it, but consult your teacher regarding this decision.

- **Project Submission**
  - There are **APAI exam dates** reserved for project grading: two in the summer, one in the autumn, and one in the winter session.
  - In practice, the team can submit their project at any time. If they accept their score, the teacher will register it **in the next available exam session**.
  - Once the project is submitted, the teacher will confirm receipt, and the review process will begin.
    - Depending on the number of submissions and the complexity of the work, the review may take some time (1 week to 2 weeks).

FAQ

1) Can we use papers from any venue (e.g., any conference, journals, arXiv), or should we choose based on specific criteria?
- You can generally use papers from any venue. However, you must first communicate with the teacher, who will either approve your choice or ask you to select a different one.

2) Can we contact the teacher for feedback while working on the project and writing the paper?
- Seeking continuous feedback will not be possible, as this is a student project rather than a collaborative effort between students and the teacher.
- In case it is really needed then there will be a reflection to your scores, as it can be interpreted as the research was done not by you but by the teacher.

3) Should we simply submit the topic name and dataset, or is a brief proposal required?
- A one-paragraph summary (about 10 lines) will be much better to allow the teacher to better understand the concept, thus supply a better feedback.